WORKED SOLUTION

# STEP II 1991, Question 13

Mechanics • Rigid Body Rotation

**STEP II 1991, Question 13**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q13

## The key idea

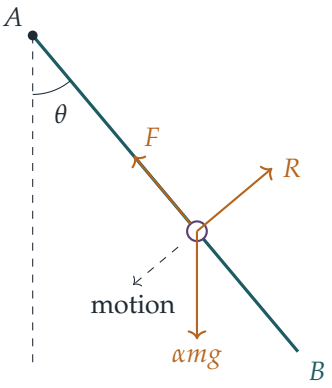
Until the ring slips, the rod and ring move together as a single rigid body pivoted at $A$. So we can find the angular speed (from energy) and the angular acceleration (from moments) of the combined body. We then use them to work out what forces the rod must exert on the ring to keep it moving with the rod. The ring stays put only while the friction it needs is at most $\mu$ times the normal reaction, and slipping starts at the first angle where that fails.

## Solution

### The model

We treat the ring as a particle at distance $x$ from $A$, the pivot as smooth, and the rod as a rigid body. Let $\theta$ be the angle between the rod and the downward vertical, so $\theta = \frac{\pi}{2}$ at release, and $\theta$ decreases as the rod swings down. Write $\omega = |\dot{\theta}|$ for the angular speed and $\beta = |\ddot{\theta}|$ for the size of the angular acceleration.

The rod exerts two forces on the ring: a normal reaction $R$ perpendicular to the rod, taken positive in the direction the ring is moving, and a friction force $F$ along the rod, taken positive towards $A$. Because the ring is threaded on the rod, $R$ may come out with either sign.




In the diagram, $R$ is drawn opposing the motion because that is how it turns out when $k^2 > xd$, but we keep the signs general in the algebra.

#### Motion of the rigid body

With the ring fixed on the rod, the moment of inertia about $A$ is $mk^2 + \alpha m x^2 = m(k^2 + \alpha x^2)$. From release, the centre of mass of the rod has fallen $d\cos\theta$ and the ring has fallen $x\cos\theta$, so conservation of energy gives

$$\tfrac{1}{2}m(k^2 + \alpha x^2)\,\omega^2 = mg(d + \alpha x)\cos\theta \quad \Longrightarrow \quad \omega^2 = \frac{2g(d + \alpha x)\cos\theta}{k^2 + \alpha x^2}.$$

Taking moments about $A$ (the pivot force has no moment), the weights have total moment $mg(d + \alpha x)\sin\theta$, so

$$\beta = \frac{g(d + \alpha x)\sin\theta}{k^2 + \alpha x^2}.$$

#### Forces on the ring

The ring moves on a circle of radius $x$. Its acceleration is $x\omega^2$ towards $A$ along the rod and $x\beta$ perpendicular to the rod in the direction of motion. The outward direction along the rod makes angle $\theta$ with the downward vertical, so the weight has component $\alpha mg\cos\theta$ outward along the rod and $\alpha mg\sin\theta$ in the direction of motion.

Along the rod, towards $A$: $F - \alpha mg\cos\theta = \alpha m x\omega^2$, so

$$\begin{aligned} F &= \alpha mg\cos\theta\left(1 + \frac{2x(d + \alpha x)}{k^2 + \alpha x^2}\right) \\ &= \frac{\alpha mg\cos\theta\,(k^2 + 3\alpha x^2 + 2xd)}{k^2 + \alpha x^2}. \end{aligned}$$

Perpendicular to the rod, in the direction of motion:

$$R + \alpha mg\sin\theta = \alpha m x\beta \quad \Longrightarrow \quad R = \alpha mg\sin\theta\left(\frac{x(d + \alpha x)}{k^2 + \alpha x^2} - 1\right) = \frac{\alpha mg\sin\theta\,(xd - k^2)}{k^2 + \alpha x^2}.$$

Notice that the $\alpha x^2$ terms cancel in $R$, which is why only $k^2 - xd$ appears.

**Ask yourself** Which way does the ring try to slide, and what does the sign of $R$ mean?

For $0 < \theta < \frac{\pi}{2}$ we have $F > 0$: the rod must pull the ring *towards* $A$, so without enough friction the ring slides outwards, towards $B$. If $k^2 > xd$ then $R < 0$, so the rod is holding the ring back. On its own the ring would fall faster than this point of the rod does, and the rod has to push against it.

#### When does it slip?

The ring stays fixed on the rod as long as $|F| \le \mu|R|$. With $k^2 > xd$, both $F$ and $|R|$ have the same positive factor $\alpha mg/(k^2 + \alpha x^2)$, so the condition is

$$\cos\theta\,(k^2 + 3\alpha x^2 + 2xd) \le \mu \sin\theta\,(k^2 - xd), \quad \text{that is,} \quad \mu \tan\theta \ge \frac{3\alpha x^2 + k^2 + 2xd}{k^2 - xd}.$$

At release, $\theta = \frac{\pi}{2}$ and $F = 0$, so the ring does not slip immediately. As $\theta$ decreases from $\frac{\pi}{2}$ to 0, $\mu\tan\theta$ decreases steadily from $+\infty$ to 0, while the right-hand side is a fixed positive number. So the inequality first fails, and the ring starts to slide outwards, at the unique angle where

$$\boxed{\mu \tan\theta = \frac{3\alpha x^2 + k^2 + 2xd}{k^2 - xd}}$$

as required. The ring always slips before the rod reaches the vertical.

#### Part (i): $k^2 = xd$

Now $R = 0$ at every angle, so the maximum available friction $\mu|R|$ is zero. But $F > 0$ as soon as $\theta < \frac{\pi}{2}$. So the ring starts sliding outwards along the rod *as soon as the rod is released*, whatever the value of $\mu$. Physically, at release the point of the rod at distance $x = k^2/d$ from $A$ starts to fall with acceleration exactly $g$ (since $x\beta = g$ at $\theta = \frac{\pi}{2}$). So the ring presses on neither side of the rod and there is nothing for friction to act through.

#### Part (ii): $k^2 < xd$

Now $R > 0$: the rod pushes the ring along in the direction of motion, because this point of the rod would fall faster than $g$ at release. The ring again tends to slide outwards, and the condition $F \le \mu R$ becomes

$$\mu \tan\theta \ge \frac{3\alpha x^2 + k^2 + 2xd}{xd - k^2}.$$

By the same argument as before, the ring holds at first and starts to slide outwards when

$$\mu \tan\theta = \frac{3\alpha x^2 + k^2 + 2xd}{xd - k^2},$$

which is the given result with $k^2 - xd$ replaced by $|k^2 - xd|$. The only difference is that the normal reaction now acts on the other side of the ring.

## Checks

- With $k^2 > xd$, as $\mu \to \infty$ the slipping angle $\theta \to 0$: very rough rods hold the ring almost to the bottom. As $k^2 - xd \to 0$, the angle tends to $\frac{\pi}{2}$, which matches part (i).
- A uniform rod of length $2a$ has $d = a$ and $k^2 = \frac{4}{3}a^2$, so the critical position $x = k^2/d = \frac{4}{3}a$ is the centre of percussion, the point that starts off in free fall when the rod is released.
- Numerically, we integrated the rigid motion, worked out the ring's acceleration by finite differences, and recovered $F$ and $R$ to within $10^{-5}$ for random parameter values. For example, with $k^2 - xd \approx 2.87$, the ring first needed more than $\mu|R|$ at $\theta \approx 1.3270$, in agreement with the formula. Cases with $k^2 < xd$ matched the formula in (ii) in the same way.

## Common slips

- Forgetting the ring's own contribution to the moment of inertia and to the moment of the weight. The ring is part of the rigid body until it slips.
- Using the friction law with the weight's normal component instead of the actual normal reaction $R$, which here also depends on the angular acceleration.
- Missing the centripetal term $x\omega^2$ in the equation along the rod, which is where the factor 3 in $3ax^2$ comes from.

## Beyond STEP

The point at distance $k^2/d$ from the pivot is the *centre of percussion*. A blow struck there produces no impulsive reaction at the pivot, which is why a cricket bat or tennis racket has a "sweet spot". It is also the length of the simple pendulum with the same period as the compound pendulum. The method here, of finding the motion of a rigid body and then asking what internal forces are needed to sustain it, is the basis of the analysis of constraint forces in Lagrangian mechanics. For more on rigid-body rotation and the compound pendulum, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.